

Data generating mechanism

Parameter values in the DGM

Data is generated according to a parametric linear regression model.

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 D + \epsilon$$

Each variable is drawn from a known distribution as follows:

Independent variable X is drawn from a standardized normal distribution, as is the continuous covariate Z . Dichotomous covariate D is drawn from a binomial distribution with equal chances of value 0 or 1. Standardized distributions are used in order to make results easily comparable and independent of scale. The random error ϵ is drawn from a normal distribution with a mean of zero and standard deviation 0.86. The standard deviation was set to 0.86 in order to achieve the desired level of power of 80% in the baseline scenario.

$$X \sim \mathcal{N}(0, 1)$$

$$Z \sim \mathcal{N}(0, 1)$$

$$D \sim \text{Bin}(1, 0.5)$$

$$\epsilon \sim \mathcal{N}(0, 0.86)$$

Five percent (2.5% at each end of the distribution) of the data is drawn with an alternative random error to increase the occurrence of outliers. The means of the alternative errors are 3 and -3 in order to simulate data points around the cut-off value for outliers in the baseline scenario (3 standard deviations from the mean).

$$\epsilon \sim \mathcal{N}(3, 1) \epsilon \sim \mathcal{N}(-3, 1)$$

The alternative outcome variable Y_a is simulated using the `faux` package. The variable is drawn from a normal distribution with a correlation of 0.6 with the outcome variable Y . This correlation was chosen in order to have strongly correlated variables but not so strong as to risk multicollinearity.

$$Y_a \sim Y, \quad \rho(Y, Y_a) = 0.6$$

The following values are held constant over the course of this project:

$$\begin{aligned} \text{iterations} &= 1600, \\ n &= 200, \\ \beta_0 &= 0, \\ \beta_1 &= 0/0.2, \\ \beta_z &= 0.06, \\ \beta_d &= 0.06 \end{aligned}$$

The β_1 parameter is the only parameter which changes between the two scenarios. In the effect scenario the parameter has a value of 0.2 and in the no-effect scenario the parameter has a value of 0. The number of iterations is based on the desired precision of the project. A Monte-Carlo Error of 1% is deemed appropriate. According to the formula by Morris et al.,

$$\begin{aligned} &\sqrt{\frac{\widehat{\text{Power}}(1 - \widehat{\text{Power}})}{n_{\text{sim}}}}, \\ 0.01 &= \sqrt{\frac{\widehat{0.8}(1 - \widehat{0.8})}{n_{\text{sim}}}}, \\ n_{\text{sim}} &= \frac{0.8 \times 0.2}{(0.01)^2}, \end{aligned}$$

the required number of iterations to achieve the desired MCE is 1600.